

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA02

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:100

Annexure A

PSEUDO CODE WRITING

Code of Conduct:

(192511154)

I M. Nishanth (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Nishanth.

Annexure A

PSEUDO CODE WRITING

Q1. Given pseudocode:

```
#define DEBUG 0
```

```
IF DEBUG
```

```
    PRINT intermediate_value
```

```
END IF
```

- Identify error in directive usage
- Correct the logic

Q2. #define PI _____

```
DECLARE radius AS float
```

```
DECLARE area AS float
```

```
INPUT radius
```

```
area = _____
```

```
PRINT area
```

Fill:

- Constant
- Formula

Q3. #define RATE 1.5

```
DECLARE x AS int = 10
```

```
DECLARE result AS float
```

```
result = x * RATE
```

```
PRINT result
```

Questions:

- Output?
- Why is the result float?

Q4. #define MAX 1000

DECLARE temp AS int
INPUT temp

IF temp > MAX
PRINT "Valid"
ELSE
PRINT "Invalid"

- Identify logical error
- Suggest correct condition

Q5. Given unordered pseudocode steps:

- DEFINE conversion rate
- PRINT result
- INPUT amount
- CALCULATE converted value

Arrange in correct order

Q6. #define MIN 10
#define MAX 50

DECLARE value AS int
INPUT value

_____ (value >= MIN AND value <= MAX)
PRINT "Within range"

_____ PRINT "Out of range"

Fill missing control structures

Q7. A billing system includes optional features such as discount calculation and tax computation. Depending on the business requirement, these features may be enabled or disabled at compile time. The program should calculate the total bill amount based on input price and quantity. If discount and tax features are enabled, they should be applied using predefined constants; otherwise, they should be skipped. The system must ensure that floating-point precision is maintained during calculations.

Write pseudocode using preprocessor directives to enable/disable features, define constants for discount and tax rates, and use appropriate data types.

Q8. #define TAX 5%

DECLARE price AS int
DECLARE total AS int

total = price + (price * TAX)

PRINT total

- Identify errors
- Correct data types and constant definition

9. A temperature conversion program supports multiple modes such as Celsius to Fahrenheit and Fahrenheit to Celsius. The mode of operation should be selected at compile time using preprocessor directives. The program should use appropriate floating-point data types to ensure accuracy and define conversion formulas using symbolic constants wherever applicable. The final output should be displayed with two decimal precision.

Write pseudocode using preprocessor directives for mode selection, constants for conversion formulas, and proper data types.

10. #define SIZE 5

DECLARE arr[SIZE]
INPUT elements

- Extend pseudocode to:
 - Find sum
 - Compute average

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Assessment Tool 1

1.
=

```
#define DEBUG 1
START
#ifdef DEBUG
    PRINT intermediate_value
#endif
STOP.
```

2.
=

```
#define PI 3.14
START
DECLARE radius AS float
DECLARE AREA AS float
Input radius
area = PI x radius x radius
PRINT area
STOP.
```

3.
=

```
#define RATE 1.5
START
DECLARE x AS int = 10
DECLARE result AS float
result = x * RATE
```

PRINT result

STOP.

4. # define MAX 1000

START

DECLARE temp AS int

INPUT temp

If temp $\leq$ Max then

printf "valid"

ELSE

printf "Invalid"

ENDIF

STOP

5. START

DEFINE Conversion_rate

Input amount

Converted_value = amount $\times$ Conversion_rate

Display Converted_value

STOP.

START
DEFINE MIN = 10
DEFINE MAX = 50
DECLARE Value AS INTEGER
READ Value
IF Value $\geq$ MIN AND Value $\leq$ MAX THEN
 Display "within range"
ELSE
 Display "out of range"
END IF
STOP.

7.
=

START
DEFINE DISCOUNT = 10%
DEFINE TAX = 5%
DECLARE price AS Float
DECLARE quantity AS INTEGER
DECLARE total AS FLOAT
READ Price
READ quantity

total = price * quantity

total = total - (total * 10/100)

total = total + (total * 5/100)

DISPLAY total

STOP.

8.
= START

DEFINE TAX = 0.05

DECLARE price AS FLOAT

DECLARE total AS FLOAT

READ price

total = price + (price * TAX)

Display total

STOP.

9.
= START

DEFINE FACTOR = 1.8

DEFINE OFFSET = 32

DECLARE celsius AS FLOAT

DECLARE fahrenheit AS FLOAT

READ celsius

fahrenheit = (celsius * FACTOR) + OFFSET

DISPLAY fahrenheit

STOP.

10.
=

START

DEFINE arr[5] AS INTEGER

DECLARE SUM AS Integer = 0

DECLARE average AS FLOAT

FOR i=0 TO 4

 READ arr[i]

 sum = sum + arr[i]

END FOR

average = sum / size

Display sum

Display Average

STOP.